

Machine Learning

Introduction

Machine Learning (ML) is a branch of AI in which computer systems learn patterns from data and use those patterns to make predictions or decisions.

Supervised Learning

Supervised learning uses labelled examples. The model learns a relationship between input features and known target values. Classification and regression are common supervised learning tasks.

Unsupervised Learning

Unsupervised learning works with data that does not have target labels. Clustering is a common technique used to group similar observations.

Model Evaluation

Machine learning models must be evaluated on suitable data. Classification can use accuracy, precision, recall, and F1-score. Regression can use metrics such as mean absolute error and mean squared error.

Overfitting

Overfitting occurs when a model learns the training data too closely and performs poorly on unseen data. Techniques such as regularization, cross-validation, and simpler models can help reduce overfitting.